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A Appendix

A Proofs for GNSA as a pseudometric

Lemma A.1 (Identity)**.** *Let X be a point cloud over some space, then* $\mathsf{GNSA}(X, X) = 0$.

Proof. Let $X = \{x_1, \ldots, x_N\}$. Let $\max_{x \in X}(d(x, 0)) = D$. Then

$$\mathsf{GNSA}(X, X) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \frac{d(x_i, x_j)}{D} - \frac{d(x_i, x_j)}{D} \right|.$$

Simplifying, we get

$$\mathsf{GNSA}(X, X) = \frac{1}{N^2 D} \sum_{1 \le i,j \le N} |d(x_i, x_j) - d(x_i, x_j)| = 0.$$

□

Lemma A.2 (Symmetry)**.** *Let X, Y be two point clouds of the same size, then* $\mathsf{GNSA}(X, Y) = \mathsf{GNSA}(Y, X)$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$. Let $\max_{x \in X}(d(x, 0)) = D_X$ and $\max_{y \in Y}(d(y, 0)) = D_Y$. Then

$$\mathsf{GNSA}(X, Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y} \right|.$$

Since $|a - b| = |b - a|$,

$$\mathsf{GNSA}(X, Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \frac{d(y_i, y_j)}{D_Y} - \frac{d(x_i, x_j)}{D_X} \right|.$$

Hence, $\mathsf{GNSA}(X, Y) = \mathsf{GNSA}(Y, X)$. □

Lemma A.3 (Non-negativity)**.** *Let X, Y be two point clouds of the same size, then* $\mathsf{GNSA}(X, Y) \ge 0$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$. Let $\max_{x \in X}(d(x, 0)) = D_X$ and $\max_{y \in Y}(d(y, 0)) = D_Y$. Then

$$\mathsf{GNSA}(X, Y) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y} \right|.$$

Since, for all i, j,

$$\left| \frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y} \right| \ge 0,$$

we get $\mathsf{GNSA}(X, Y) \ge 0$. □

Lemma A.4 (Triangle inequality)**.** *Let X, Y and Z be three point clouds of the same size, then* $\mathsf{GNSA}(X, Z) \le \mathsf{GNSA}(X, Y) + \mathsf{GNSA}(Y, Z)$.

Proof. Let $X = \{x_1, \ldots, x_N\}$, $Y = \{y_1, \ldots, y_N\}$ and $Z = \{z_1, \ldots, z_N\}$. Let $\max_{x \in X}(d(x, 0)) = D_X$, $\max_{y \in Y}(d(y, 0)) = D_Y$ and $\max_{z \in Z}(d(z, 0)) = D_Z$. Then

$$\mathsf{GNSA}(X, Z) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \frac{d(x_i, x_j)}{D_X} - \frac{d(z_i, z_j)}{D_Z} \right|.$$

For each i, j, add and subtract $\frac{d(y_i, y_j)}{D_Y}$, then

$$\mathsf{GNSA}(X, Z) = \frac{1}{N^2} \sum_{1 \le i,j \le N} \left| \left(\frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y} \right) + \left(\frac{d(y_i, y_j)}{D_Y} - \frac{d(z_i, z_j)}{D_Z} \right) \right|.$$

Since $|a+b| \leq |a| + |b|$,

$$\mathsf{GNSA}(X,Z) \leq \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left(\left| \frac{d(x_i,x_j)}{D_X} - \frac{d(y_i,y_j)}{D_Y} \right| + \left| \frac{d(y_i,y_j)}{D_Y} - \frac{d(z_i,z_j)}{D_Z} \right| \right).$$

Hence,

$$\mathsf{GNSA}(X,Z) \leq \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left| \frac{d(x_i,x_j)}{D_X} - \frac{d(y_i,y_j)}{D_Y} \right| + \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left| \frac{d(y_i,y_j)}{D_Y} - \frac{d(z_i,z_j)}{D_Z} \right|,$$

or $\mathsf{GNSA}(X,Z) \leq \mathsf{GNSA}(X,Y) + \mathsf{GNSA}(Y,Z)$. □

From Lemma A.1, Lemma A.2, Lemma A.3 and Lemma A.4, we see that GNSA is a psuedometric over the space of point clouds.

B Proofs for GNSA as a similarity metric

B.1 Invariance to Isotropic Scaling

Lemma B.1 (Invariance to Isotropic scaling in the first coordinate)**.** *Let X and Y be two point clouds of the same size. Let $c \in \mathbf{R}$ and $c \neq 0$, X_c be the point cloud with each point in X scaled by a factor of c, then* $\mathsf{GNSA}(X,Y) = \mathsf{GNSA}(X_c,Y)$.

Proof. Let $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$, then $X_c = \{cx_1, \ldots, cx_N\}$. Let $\max_{x \in X}(d(x,0)) = D_X$, $\max_{y \in Y}(d(y,0)) = D_Y$ and $\max_{x \in X_c}(d(x,0)) = D_{X_c}$. Since, each point in X_c is the c times each point in X. Then, we can write $D_{X_c} = \max_{x \in X}(d(cx,0))$. Since, for any two points, x_1 and x_2, $d(cx_1, cx_2) = |c| d(x_1,x_2)$, then $D_{X_c} = |c| \max_{x \in X}(d(x,0))$. Hence, $D_{x_c} = |c| D_X$. From the definition of GNSA,

$$\mathsf{GNSA}(X_c,Y) = \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left| \left(\frac{d(cx_i,cx_j)}{D_{X_c}} - \frac{d(y_i,y_j)}{D_Y} \right) \right|.$$

Again, using $d(cx_i,cx_j) = |c| d(x_i,x_j)$ and $D_{X_c} = |c| D_X$,

$$\mathsf{GNSA}(X_c,Y) = \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left| \left(\frac{|c| d(x_i,x_j)}{|c| D_X} - \frac{d(y_i,y_j)}{D_Y} \right) \right|.$$

Simplifying,

$$\mathsf{GNSA}(X_c,Y) = \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left| \left(\frac{d(x_i,x_j)}{D_X} - \frac{d(y_i,y_j)}{D_Y} \right) \right|.$$

Hence, $\mathsf{GNSA}(X_c,Y) = \mathsf{GNSA}(X,Y)$. □

By Lemma A.2, we can see that this gives us invariance to isotropic scaling in the second coordinate as well.

Lemma B.2 (Invariance to Isotropic scaling in the second coordinate)**.** *Let X and Y be two point clouds of the same size. Let $c \in \mathbf{R}$ and $c \neq 0$, Y_c be the point cloud with each point in Y scaled by a factor of c, then* $\mathsf{GNSA}(X,Y) = \mathsf{GNSA}(X,Y_c)$.

Combining Lemma B.1 and $Lemma\ B.2$, we get the required lemma.

Lemma B.3 (Invariance to Isotropic scaling)**.** *Let X and Y be two point clouds of the same size. Let $c_1, c_2 \in \mathbf{R}$, $c_1 \neq 0$ and $c_2 \neq 0$, X_{c_1} be the point cloud with each point in X scaled by a factor of c_1 and Y_{c_2} be the point cloud with each point in Y scaled by a factor of c_2, then* $\mathsf{GNSA}(X,Y) = \mathsf{GNSA}(X_{c_1},Y_{c_2})$.